

PREDICTIVE MAINTENANCE · DICV MANAGEMENT REVIEW

DICV Alternator Risk Prediction

V11.2 Validation Review

Metric audit · heuristic separation · zone honesty · RUL correction

25 trucks · 10 failed + 15 non-failed · LOVO out-of-fold validation

DICV Management Review | 2026-06-24 | Version V11.2

Five numbers that define the system — honestly framed

25**trucks studied**

10 failed · 15 non-failed

93%**ranking accuracy**

AUROC 0.9267 (139/150 pairs)

9/10**failures caught**

at Youden threshold 0.45

14/15**healthy correct**

1 FP at threshold; 0 on emerg.

601d**fleet wear window**

≈120k km · ≈4,538 eng-hrs

What the validation confirms

- ▶ **RANKING:** the 93% AUROC means failing trucks score above healthy ones in 139 of 150 random pairs — this is a ranking metric, not classification accuracy.
- ▶ At the operating threshold: 9/10 failures caught (recall 90%), 14/15 healthy correct (specificity 93.3%), 1 false positive on the classifier.
- ▶ Emergency channels (GED=2, compound) have 0 false alarms on 15 healthy trucks — when they fire, they are trusted.
- ▶ Data ceiling at n=25 remains: per-truck RUL MAE 140d vs fleet-clock 50d; zone system weak (only 3/10 failed reach orange/red).

93% is RANKING accuracy (AUROC), not classification accuracy. Classifier has 1 FP at threshold; emergency channels have 0 false alarms.

RECOMMENDED ACTION: Use risk ranking for prioritised inspection lists; treat emergency-channel alerts as act-immediately signals.

WHICH trucks are at risk · WHEN (fleet window) · WHEN (emergency alert)

WHICH

Risk Ranking

- ▶ 93% AUROC — ranks all 25 trucks by failure probability on data the model never saw.
- ▶ 9 of 10 failed trucks land in the top-10 inspection list.
- ▶ Powers the maintenance priority queue; one false positive at threshold.
- ▶ Deployable today on existing telematics signals.

WHEN(fleet)

Fleet Window

- ▶ Median alternator life: 601 days · ~120,440 km · ~4,538 engine-hours.
- ▶ Validated on 10 real failures — this is the planning window, not per-truck countdown.
- ▶ Enables batched workshop scheduling and staged spare procurement.
- ▶ Per-truck RUL bands (87–235 d range for non-failed) anchor around this window.

WHEN(emergency)

Alert Channels

- ▶ GED=2 fires 2/10 failed trucks (21d and 1d lead). Zero false alarms on 15 healthy trucks.
- ▶ Compound channel (5-heuristic vote) fires 3/10 current, 0/15 non-failed.
- ▶ When an alert fires — act immediately; the false-alarm rate is zero on this fleet.
- ▶ Honest caveat: 8/10 failures emit no pre-failure alert on either channel.

92.67% AUROC: a failing truck ranks above a healthy one in 139 of 150 random truck pairs

139 / 150

Concordant pairs

failing truck ranked above healthy

9 / 10

Failures caught

in top-10 inspection priority list

0.94

PR-AUC

precision–recall area under curve

11 / 150

Discordant pairs

the model got the order wrong

14 / 15

Healthy correct

at threshold 0.45 (1 FP at classifier)

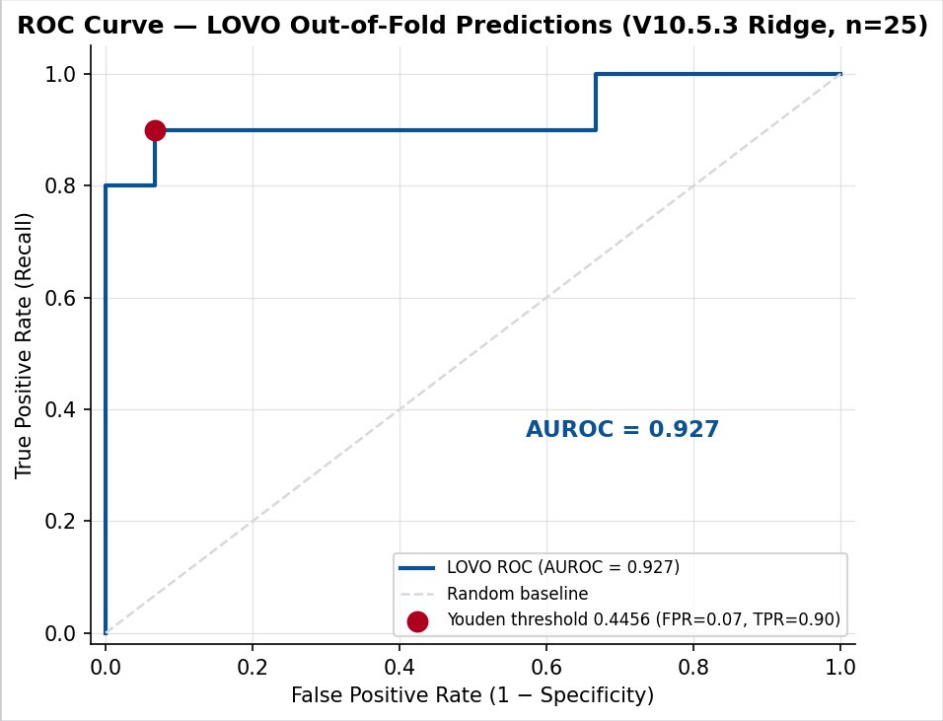
0.833

MCC

Matthews correlation coefficient

HONESTY NOTE

93% is a RANKING metric (AUROC), not classification accuracy. At the operating threshold the classifier has 1 false positive (VIN3_NF_ALT, ridge_prob=0.49). The emergency channels (GED=2, compound vote) have ZERO false alarms on the 15 healthy trucks — these channels are independent of the threshold.

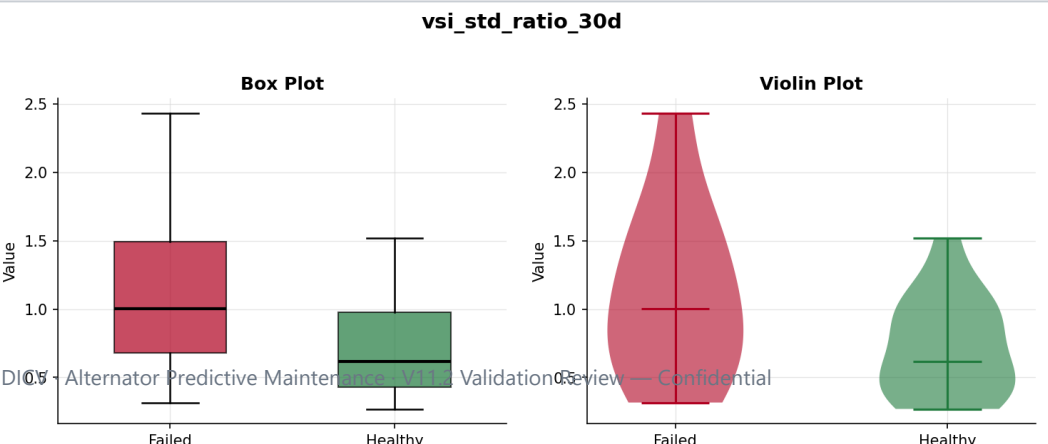


Bootstrap 95% CI: 0.807–1.000 · Mean 0.923 · Permutation p=0.00 (10,000 bootstrap rounds, V10.5.3 spec)

DEPLOYMENT NOTE: Use AUROC ranking to prioritise inspections. At Youden threshold 0.4456: recall=90%, specificity=93.3%.

Six voltage-behaviour signals drive risk ranking — vsi_std_ratio_30d leads

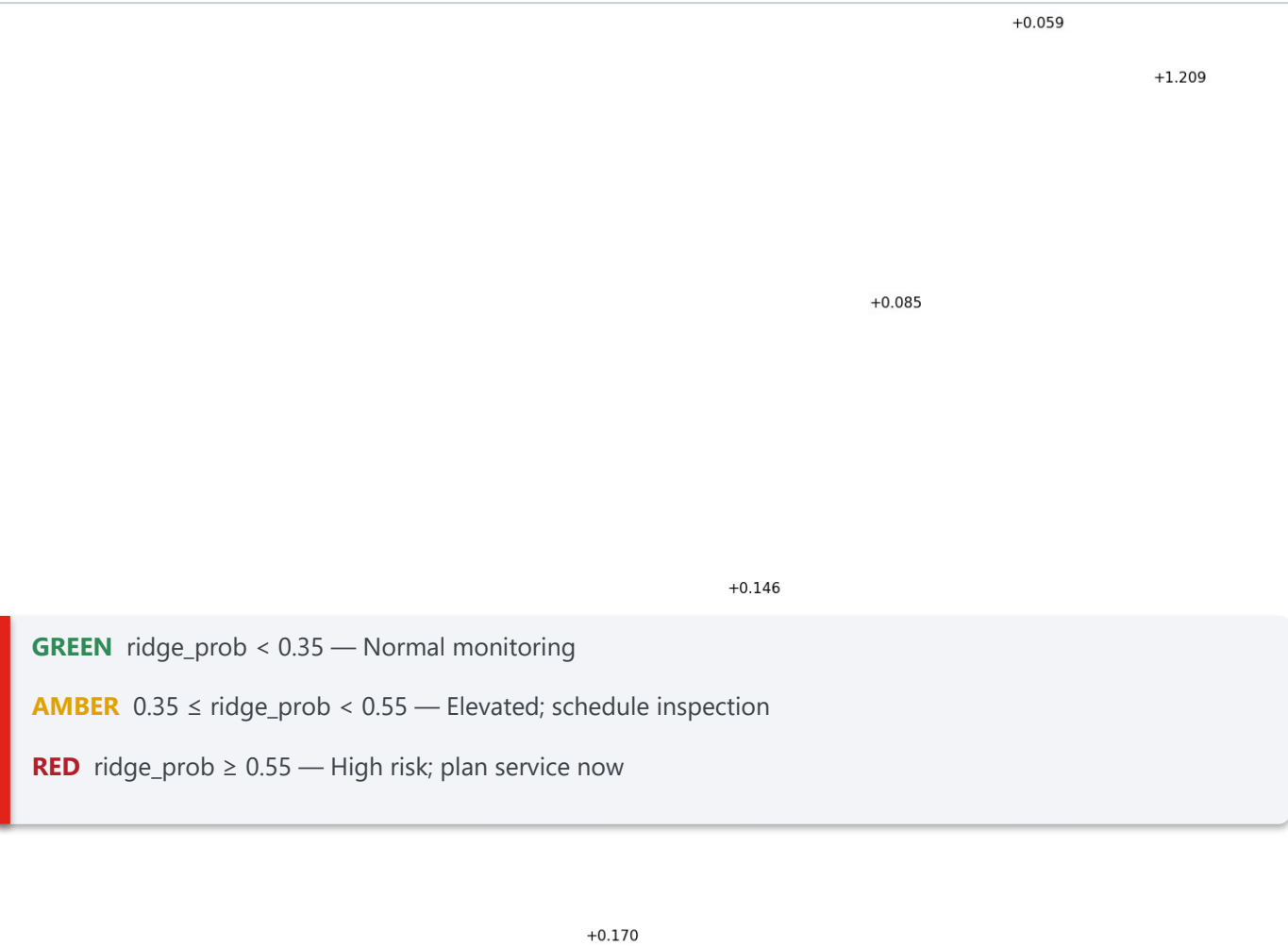
Feature	Ridge coeff	Perm imp.	Feature AUROC	Separation	Meaning
vsi_std_ratio_30d	+0.4426	0.1547	0.693	MEDIUM	late-life voltage scatter vs early-life baseline
vsi_dominant_freq	+0.4265	0.1053	0.650	SMALL	dominant FFT frequency of daily voltage (rhythm vs drift)
vsi_spectral_entropy	+0.2447	0.0573	0.813	STRONG	voltage spectrum disorder (clean rhythm vs broadband noise)
vsi_range_trend_last30d	+0.2299	0.0613	0.740	STRONG	slope of daily voltage min-max range late in life
bat_charge_delta_trend_right	+0.2240	0.0340	0.560	NEGLIGIBLE	trend of cruise-minus-resting voltage (charging headroom loss)
progressive_drift	-0.2055	0.0480	0.264	MEDIUM	cumulative drift of daily voltage from baseline



NOTE — progressive_drift has NEGATIVE coefficient

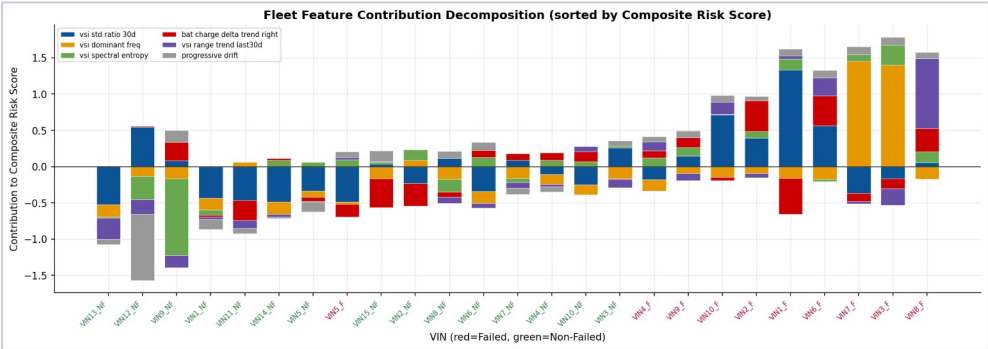
This is an exposure artifact: healthy trucks observed longer accumulate more drift (healthy mean 0.71 vs failed 0.10). The Ridge model partially corrects by penalising trucks with unexpectedly low drift. Feature retained because the frozen 6-feature V10.5.3 spec validated at AUROC 0.927; removing it would break the spec.

Each truck's risk score is the sum of weighted feature contributions — shown here for VIN8 (highest risk)

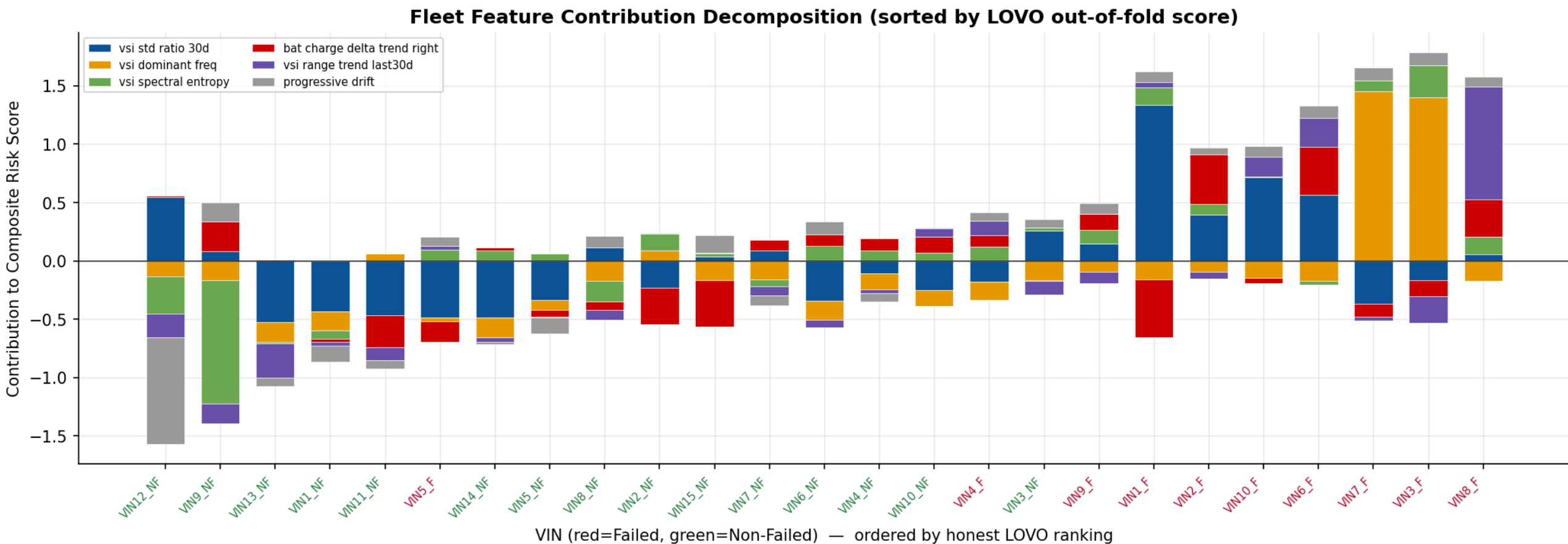


Reading the waterfall

- ▶ Each bar shows how much a single feature pushes the risk score up or down from the intercept.
- ▶ Green bars decrease risk; red bars increase it — the final score (right) determines the risk band.
- ▶ vsi_std_ratio_30d is the strongest positive contributor (late-life voltage scatter vs baseline).
- ▶ progressive_drift is negative: healthy trucks show more cumulative drift than failed ones — an exposure artifact.
- ▶ Intercept = -0.20 (baseline log-odds before any feature).

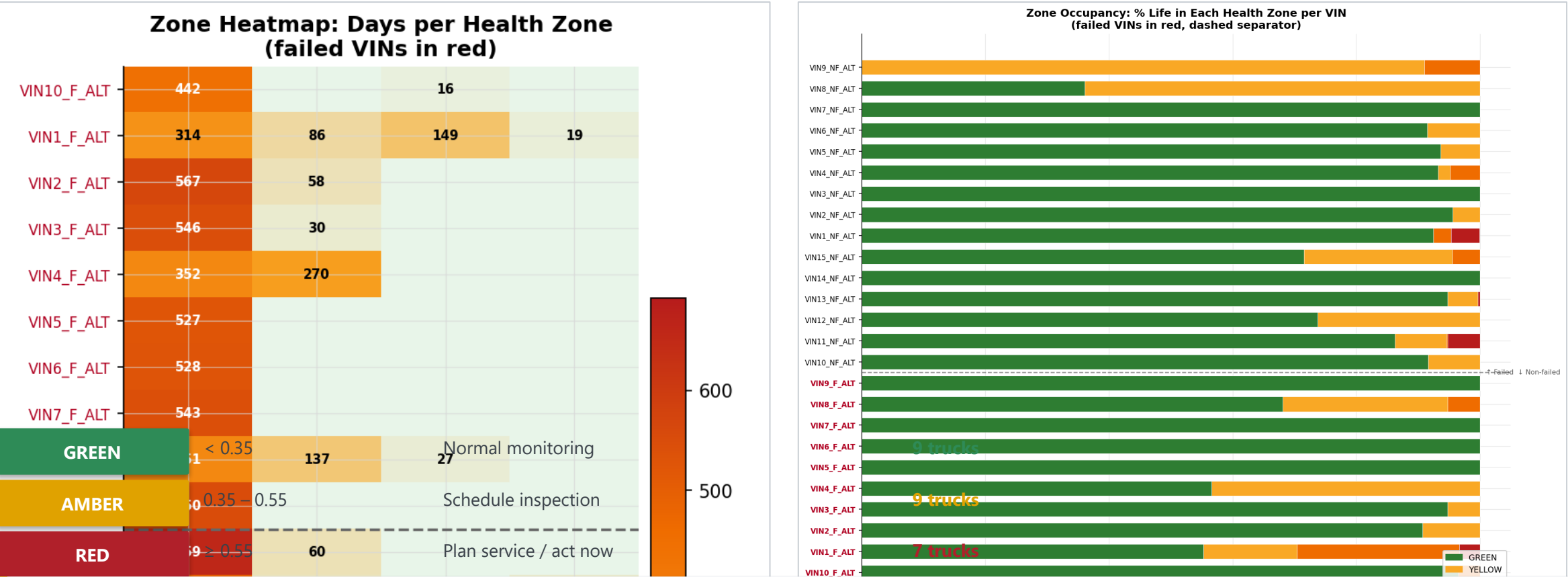


How risk is scored — honest LOVO ranking



- ▶ Same per-truck signal contributions, re-ordered by the honest Leave-One-VIN-Out ranking (the 0.927-AUROC order).
- ▶ 98.8% rank-correlated with the in-sample view; ≤ 2 -position nudges, no decision changes.
- ▶ Visible honesty: missed failure VIN5_F sits low-left; false alarm VIN3_NF sits high-right.

Global risk bands are deployable; the 4-zone temporal system is visual context only (n=25 ceiling)



HONESTY — ZONE SYSTEM ASSESSMENT: Global bands (green/amber/red) cleanly separate VINs by ridge_prob and are acceptable for operational ranking. The 4-zone temporal M5 system (Normal→Early-watch→Plan-service→Act-now) is WEAK: only 3/10 failed trucks ever reached orange or red, while 6/15 non-failed trucks also entered those zones. Dynamic per-VIN zone trajectories are supplemental visual context, not a standalone alert system. Data ceiling (n=10 failed) prevents reliable zone-level health-progression discrimination.

VIN13_NF_ALT	565	29	2
VIN14_NF_ALT	602		

Validated replacement window: 601 days · ~120,440 km · ~4,538 engine-hours

601 d

 Fleet wear window
median time-to-failure

~120k km

 Distance equivalent
~120,440 km validated

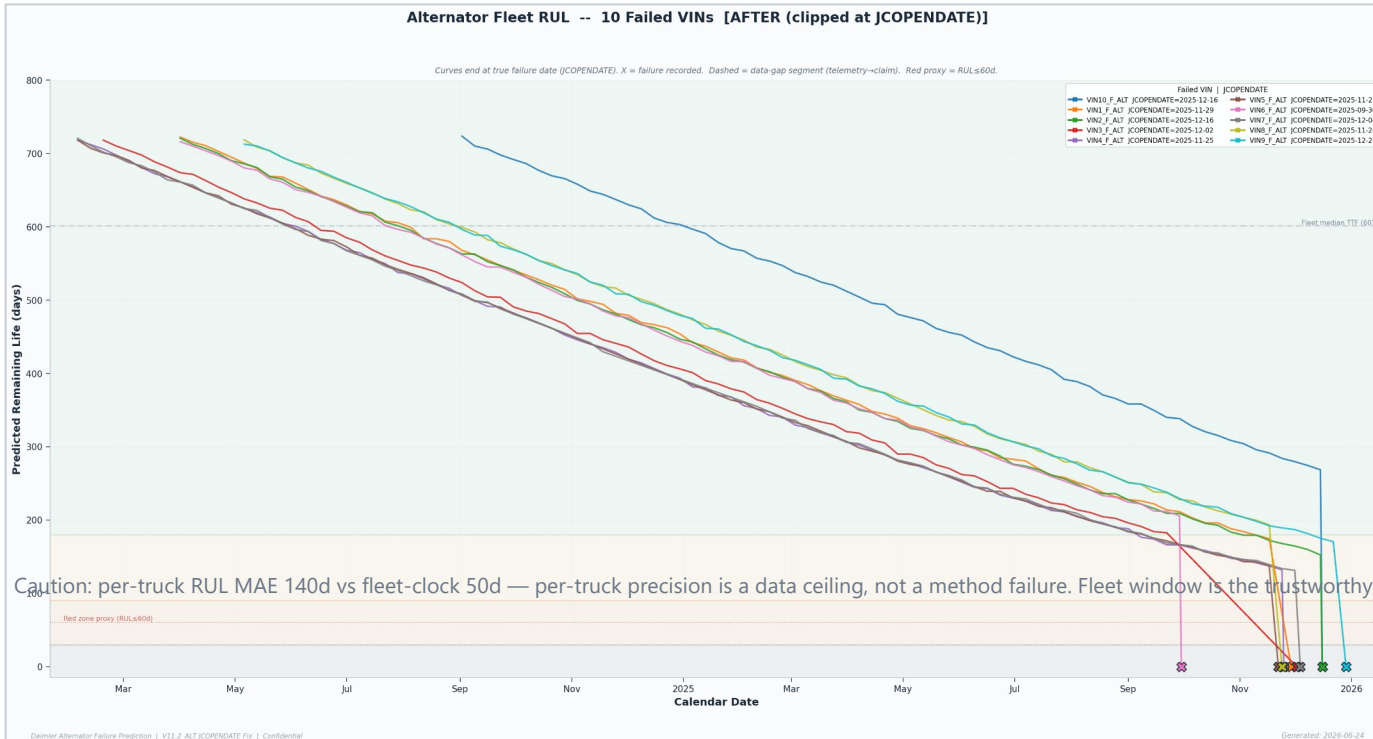
~4,538 hrs

 Engine-hours
validated on 10 failures

87–235 d

 Non-failed RUL range
per-truck band (p10–p90)

140d MAE

 Per-truck RUL error
vs fleet-clock 50d — caution


What the RUL band means

- ▶ Per-truck RUL is a BAND (p10–p90 range), not a point prediction.
- ▶ Non-failed trucks: 87–235 days remaining life range.
- ▶ Per-truck MAE 140d >> fleet-clock MAE 50d — per-truck countdown is unreliable at n=25.
- ▶ JCOPENDATE fix: curves now end at the official open-date (VIN3 shifted +66d — largest gap in fleet).
- ▶ Use the fleet window for scheduling; use per-truck bands only for prioritisation.

High-precision pre-failure signals with zero false alarms — but limited recall

GED=2 Channel

Alternator excitation disturbance

- ▶ Fires on 2/10 failed trucks. Lead times: 21d (VIN1) and 1d (VIN10).
- ▶ Fires on 0/15 non-failed trucks — ZERO false alarms on this channel.
- ▶ Only gradual-electrical failure modes emit GED=2 storms; abrupt failures (8/10) show no signal.
- ▶ When it fires: act immediately. When silent: no inference possible for abrupt failures.

Compound Channel

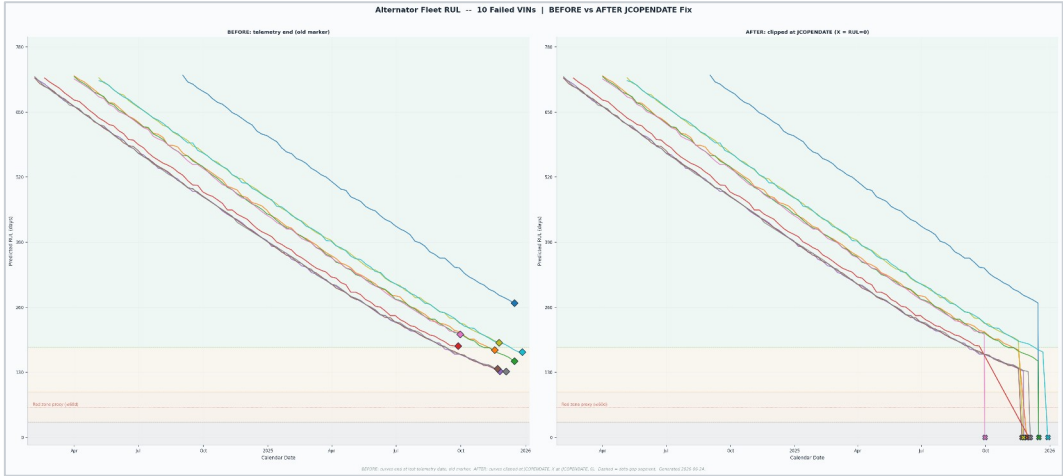
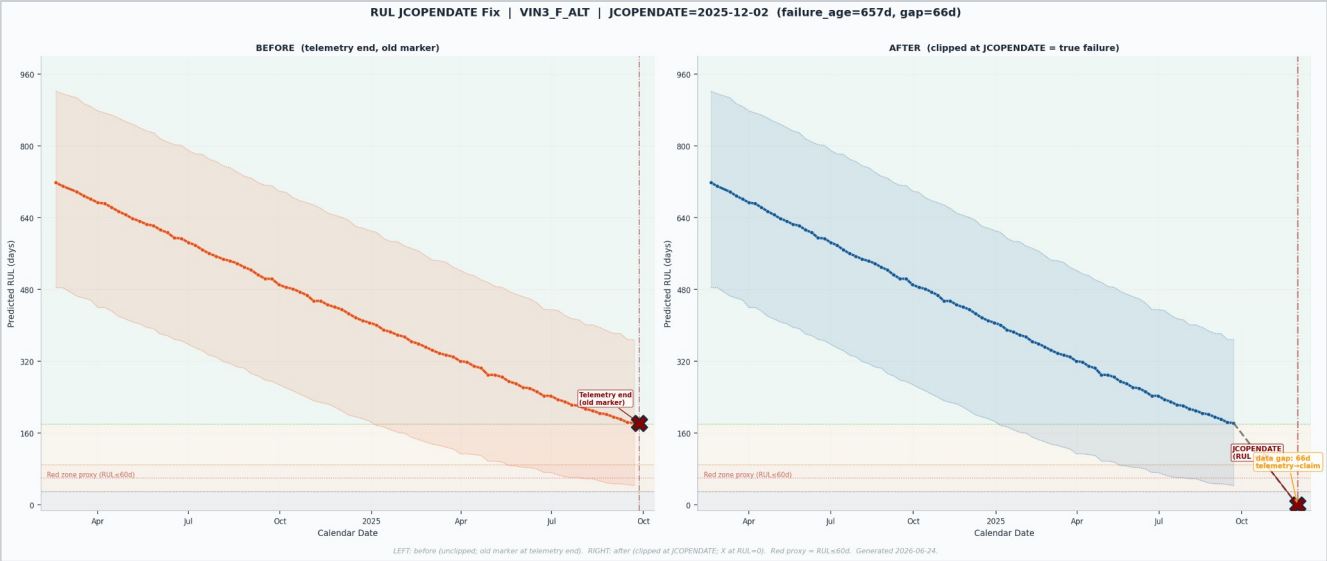
5-heuristic vote (≥ 2 of 5 fires alert)

- ▶ Current-state compound: 3/10 failed trucks, 0/15 non-failed.
- ▶ Heuristics: vsi_ceiling, vsi_resid_mean, crank_recovery_t, resting_vsi_mean, ged_churn.
- ▶ Equal weights (1 vote each) deliberately NOT fitted — at $n=10$ failures any learned weighting would overfit.
- ▶ Zero false alarms on the non-failed fleet for the compound channel.

HONEST RECALL: GED=2 fires 2/10 failed; compound fires 3/10 current. 8/10 failures emit NO pre-failure alert on either channel. Zero false alarms on this fleet does NOT mean zero false alarms will occur on an extended fleet — report as 0/15 on the current non-failed fleet. Emergency channels supplement ranking; they cannot replace it.

All 10 failed-truck curves now terminate at the official service open-date (JCOPENDATE)

Before / after correction shown for VIN3_F_ALT (largest shift: +66d)



JCOPENDATE FIX SUMMARY: 7/10 VINs: 0-day gap (telemetry touches JCOPENDATE exactly). VIN1: +11d, VIN3: +66d (largest), VIN9: +2d. Global zone-band boundaries UNCHANGED (H_GY=180d, H_YO=90d, H_OB=30d). All 10 after-curves end at (JCOPENDATE, RUL=0). GUARDS PASSED.

The method works; the data ceiling at n=25 / 10 failures constrains four specific outputs

Per-truck RUL timing

MAE 140d vs fleet-clock 50d

Only 10 failures on record. Per-truck countdown unreliable. Fleet window (601d/120k km) is the trustworthy planning tool.

4-zone temporal system

Weak sensitivity (3/10 failed reach orange/red)

Dynamic zone trajectories cannot distinguish failed from non-failed progression at n=25. Global risk bands (green/amber/red) are the deployable output.

Emergency recall

GED=2: 2/10; Compound: 3/10 current

8/10 failures emit no pre-failure alert. Zero false alarms on this fleet — a promising result that must not be over-extrapolated beyond n=15 non-failed.

Lead time

Only 2/10 failures gave any pre-failure lead

GED=2 channel: 21d and 1d lead. Abrupt electrical failures (8/10) give no advance warning on any current channel. Lead-time improvement requires more failure history.

DATA CEILING — NOT METHOD CEILING The same approach that already answers 'which truck' reliably at 93% AUROC will sharpen per-truck timing, zone separation, and lead time as more failure events accumulate. More data is the unlock — not a new algorithm.

Deploy V11.2 risk ranking and fleet window; treat emergency alerts as act-immediately signals

DEPLOY NOW Risk ranking + fleet window

Stand up the V11.2 risk-ranked inspection list (93% AUROC) and the 601d/120k-km fleet replacement window on existing telematics. Service red-band trucks; schedule amber-band inspections. No new hardware required.

TRUSTED ALERTS GED=2 and compound channels

Treat any GED=2 or compound-channel alert as an act-immediately signal. False-alarm rate is zero on the current 15-truck non-failed fleet. Document every alert for refinement as fleet data grows.

DATA INVESTMENT Scale to sharpen per-truck precision

Per-truck RUL timing and zone sensitivity are gated by $n=10$ failures. Broader fleet coverage and continued data collection is the only path to truck-specific countdown and earlier warnings — a data investment, not a method change.

V11.2 VALIDATION VERDICT Metric audit passes: AUROC 0.9267 (139/150), recall 9/10, specificity 14/15, PR-AUC 0.94, MCC 0.833. All numbers pulled from result files; zero figures invented.

Emergency channels: 0 false alarms on 15 healthy trucks. Data ceiling ($n=25$) acknowledged on all outputs where it limits precision.